

AI-Powered Document Intelligence

THE 5-STEP PIPELINE ARCHITECTURE



Stage & Ingest

Secure landing zone with auto-tracking directory for all incoming PDFs.

AI_PARSE_DOCUMENT

Layout-aware extraction.
Medical tables and structures fully preserved.

Cortex Search

Semantic indexing for meaning-based clinical concept retrieval.

Snowflake CoWork

Conversational reasoning interface delivering accurate, cited answers.

Scheduled Task

Native automation for hands-free, incremental document ingestion.



Clone the notebook:

github.com/sfc-gh-aprah/AI-Powered-Document-Intelligence-for-Healthcare

Step 1: Stage & Ingest

Everything starts with a **Stage**—a governed landing zone inside your security perimeter. We load 6 real-world style clinical documents representing two patient journeys.

Anna Doe (HF)

66-year-old with heart failure. Fragmented data sources:

- Lab Report (Lab System)
- Hospital Discharge Summary (Inpatient EHR)
- Cardiac Rehab Prior Auth (Payer Form)
- Follow-up Progress Note (Outpatient Clinic)

Thomas Roe (TKA)

53-year-old needing a knee replacement:

- Prior Auth Request (Documenting failed conservative therapies)
- Surgical Discharge Summary (Post-op course)

Step 2: AI_PARSE_DOCUMENT



Layout Mode

Not just OCR text-scraping. Clinical documents live and die by structure. Medication tables, lab panels, and diagnosis lists come out intact as queryable markdown tables.



Massive Scale

AI_PARSE_DOCUMENT natively handles documents up to 2,000 pages in a single call. Full longitudinal charts or claims files require no pre-splitting.



Embedded Images

In preview today: Extract signatures and clinical figures directly from PDFs and Word docs to power advanced compliance and verification workflows.

Structure Preserved: Anna's Medication Table

Extracted perfectly from the PDF via AI_PARSE_DOCUMENT into structured markdown. No regex required.

Medication	Dose	Frequency	Instructions
Furosemide	60 mg PO	BID	Take with food; weigh daily
Lisinopril	20 mg PO	Daily	Hold if SBP < 90
Carvedilol	12.5 mg PO	BID	Take with meals
Metformin	1000 mg PO	BID	Take with meals

AI_EXTRACT: Confidence Scores

98%

Confidence Score

Targeted Field Extraction

Using AI_EXTRACT on Thomas Roe's faxed prior auth form with zero template configuration.

CPT Code: 27447 (Total Knee Arthroplasty)

ICD-10 Dx: M17.11 (Primary OA, right knee)

Est. Cost: \$32,450.00

Every value returns a confidence score. This enables human-in-the-loop routing: auto-approve high confidence, flag edge cases for review.

Step 3: Cortex Semantic Search



The Keyword Problem: Clinicians don't write WHERE clauses. They ask questions in clinical language that rarely matches the exact document wording.



Search by Meaning: If you search "patient with fluid overload", semantic search understands that connects directly to "acute decompensated heart failure" in Anna Doe's summary.



Zero Maintenance: One DDL statement creates the service. No embedding models to pick, no vector databases to stand up.



Traceable Origins: File name and page number travel with every result, ensuring answers trace back to the exact source page.

Step 4: Snowflake CoWork

Conversational Clinical Intelligence

Transition from simply searching documents to actively querying an intelligent agent. Snowflake CoWork provides a natural language interface that reasons over your governed data.

Strict Behavioral Instructions

By enforcing clear guidelines, agents synthesize complex patient histories safely:

"Always cite the source document and page for every clinical fact. If information is not in the documents, say so. Never speculate about patient information."



600 × 400

Synthesizing Answers via Snowflake CoWork

Thomas Roe (Prior Auth)

Q: "What conservative treatments did he fail?"

Agent: 16 PT sessions. Meloxicam discontinued for GI side effects. Steroid injection (<3 weeks relief). Synvisc series.

[Cited to Prior Auth Form]

Anna Doe (Progress)

Q: "What is her most recent kidney function?"

Agent: Creatinine improved from 1.6 on admission to 1.4 at her follow-up.

[Synthesized across Discharge Summary and Progress Note]

Step 5: Native Automation

Incremental by construction. New files in, new rows out.



Scheduled Tasks Run on a schedule, only parsing net-new files from the stage.



Target Lag Refresh Cortex Search keeps the index refreshed automatically. No glue code required.

The Snowflake Reality

Concern Area	The Snowflake Answer
Compute Cost	Parsing is billed per page. The pipeline, index, and query architecture runs efficiently on standard Small warehouses.
Document Scale	AI_PARSE_DOCUMENT natively processes documents up to 2,000 pages in a single execution.
Data Privacy	PHI never leaves the boundary. Models run entirely inside Snowflake under existing RBAC policies.
Auditability	AI Agent Identity grants agents dedicated roles, providing a complete, first-class audit trail for compliance.